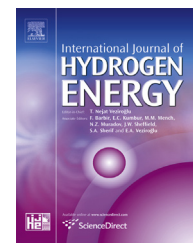


Available online at www.sciencedirect.com

ScienceDirect

journal homepage: www.elsevier.com/locate/he

Research and development of a laboratory scale Totalized Hydrogen Energy Utilization System



Satya Sekhar Bhogilla ^a, Hiroshi Ito ^a, Atsushi Kato ^b, Akihiro Nakano ^{a,*}

^a National Institute of Advanced Industrial Science and Technology, Research Institute for Energy Conservation, 1-2-1 Namiki, Tsukuba East, Tsukuba, Ibaraki 305-8564, Japan

^b R & D Center, Takasago Engineering Co., Ltd, Japan

ARTICLE INFO

Article history:

Received 11 August 2015

Received in revised form

26 October 2015

Accepted 26 October 2015

Available online 27 November 2015

Keywords:

Hydrogen system

Unitized reversible fuel cell

Hydrogen storage

Metal hydride

Energy efficiency

ABSTRACT

Hydrogen is accepted as the energy carrier for future. Hydrogen based energy storage system represents an important option to store the fluctuating power generated from the renewable sources. A laboratory scale Totalized Hydrogen Energy Utilization System (THEUS) has been developed. THEUS consists of the Unitized Reversible Fuel Cell (URFC), which has both fuel cell and water electrolysis functions in one component, and hydrogen storage tank. THEUS was tested separately in water electrolyzer mode operation and fuel cell mode operation. THEUS was also tested continuously for three days to evaluate the energy efficiency. The URFC and hydrogen storage tank performance were characterized. In particular the overall system efficiency, stability upon cycling, and the reaction heat recovery rates of the metal hydride tank were discussed.

Copyright © 2015, Hydrogen Energy Publications, LLC. Published by Elsevier Ltd. All rights reserved.

Introduction

In recent years, hydrogen-based energy conversion systems have been receiving much attention. Hydrogen is an energy carrier that can be used to stabilize the usage of intermittent renewable energy [1,2]. The focus on stand-alone renewable energy systems is highlighted by a major and extensive research effort, which has resulted in numerous experimental projects and the development of several simulation models [3–17]. A hydrogen-based energy conversion system consists of a hydrogen production unit (water electrolyzer), a storage unit, and a unit which converts hydrogen back to electricity (fuel cell). The unitized reversible fuel cell (URFC) can operate alternately as a water electrolyzer (WE) and as a fuel cell (FC).

URFCs are highly compact and they allow considerable system simplification without any system performance reduction when compared to separate WE and FC systems. Similarly when compared with conventional batteries, URFC has a higher energy density, long-term operation, durability, and environmental protection [18,19]. The concept of Totalized Hydrogen Energy Utilization System (THEUS) is shown in Fig. 1; it consists of the URFC, which has both fuel cell and water electrolysis functions in one component, and the metal hydride tank (MHT). THEUS was developed for the purpose of load leveling in commercial buildings. THEUS stores the electricity energy at night time by using a WE and generates power during the day by using FC. The system also utilizes the cold heat of reaction heat for air conditioning. THEUS was also designed for storing renewable energy such as solar energy

* Corresponding author. Tel.: +81 29 861 7250; fax: +81 29 851 7523.

E-mail address: a.nakano@aist.go.jp (A. Nakano).

<http://dx.doi.org/10.1016/j.ijhydene.2015.10.105>

0360-3199/Copyright © 2015, Hydrogen Energy Publications, LLC. Published by Elsevier Ltd. All rights reserved.

Abbreviations

ABS	Absorption
DES	Desorption
FC	Fuel cell
MH	Metal hydride
WE	Water electrolyzer
Q_{WE_HEAT}	WE-Heat output, J
Q_{WE_OUT}	WE-Hydrogen energy output, J
Q_{WE_IN}	WE-Power input, J
Q_{WE_ACCS}	WE-Input power to accessories, J
Q_{WE_TO}	WE-Total energy output, J
Q_{MH_ABS}	MH-ABS heat output, J
Q_{WE_TI}	WE-Total energy input, J
η_{WE}	WE-Efficiency, %
η_{WE+H}	WE-Efficiency including URFC heat output, %
$\eta_{WE+H+MH}$	WE-Efficiency including URFC and MH heat output, %
Q_{FC_OUT}	FC-Power output, J
Q_{FC_HEAT}	FC-Heat output, J
Q_{FC_ACCS}	FC-Input power to accessories, J
Q_{FC_IN}	FC-Hydrogen energy input, J
Q_{FC_TO}	FC-Total energy output, J
Q_{MH_DES}	MH-DES heat output, J
Q_{FC_TI}	FC-Total energy input, J
η_{FC}	FC-Efficiency, %
η_{FC+H}	FC-Efficiency including URFC heat output, %
$\eta_{FC+H+MH}$	FC-Efficiency including URFC and MH heat output, %
Q_{TO}	Total energy output, J
Q_{TI}	Total energy input, J
η_{URFC}	Total efficiency of URFC, %
η_{THEUS}	Total efficiency of THEUS, %
ε	Reaction heat recovery rate, %

and wind energy, and bridge the gap between renewable energy supply and demand. It can supply electric power, thermal energy, and hydrogen to meet energy demand. The possibility of renewable energy storage and the flexibility in using hydrogen as a fuel for FC vehicles can make THEUS attractive as an energy storage solution for all types of energy systems [20–23].

Nevertheless, storage of hydrogen is a challenging task because hydrogen is a low density gas and has a wide range of flammability. The pressurized gas storage system requires surplus energy for compression, special maintenance, and safety precautions. The cryogenic liquefaction also requires surplus energy [24,25]. Chemical hydride storage by using methylcyclohexane requires thermal energy for hydrogen generation and high temperatures restricts its use in practical applications [26]. Among the storage options, metal hydride based hydrogen storage system has stood ahead of other methods for small scale (from 10 NL to 1000 Nm³ H₂) hydrogen storage. This method uses a metal hydride alloy that can absorb and store hydrogen with relatively high volumetric density and form mainly solid metal-hydrogen compounds. Metal hydride based hydrogen storage systems deserve attention as they can store hydrogen at near ambient pressure and temperature over a long period of time. Hence, metal hydride provides a platform for hydrogen storage with a high standard of safety for stand-alone applications [27,28]. Absorption of hydrogen in metal hydrides is an exothermic reaction. Whereas, desorption of hydrogen is an endothermic reaction. Therefore, reaction heat can be utilized as a thermal output [29]. In this paper, the experimental results obtained from the THEUS based on the integration of URFC with a MHT have been discussed. Results of the MHT characterizations as well as the integrated system behavior are reported.

Experimental set-up

The lab. scale THEUS developed at National Institute of Advanced Industrial Science and Technology (AIST) is shown in Fig. 2(a) and schematic view is shown in Fig. 2(b). The URFC was manufactured in Takasago Engineering Co., Ltd, Japan. The THEUS consists of URFC, MHT, and their auxiliary components. The hydrogen flows in between the URFC and MHT. The solid arrows and dashed arrows in Fig. 2(b) show the hydrogen flow direction in WE mode and FC mode, respectively. The hydrogen flow rate was measured by a hydrogen flow controller, Kofloc Model 3665, which was used as a mass flow meter in this study. The full scale (FS) value of the hydrogen flow controller was 50 NL/min, and the accuracy

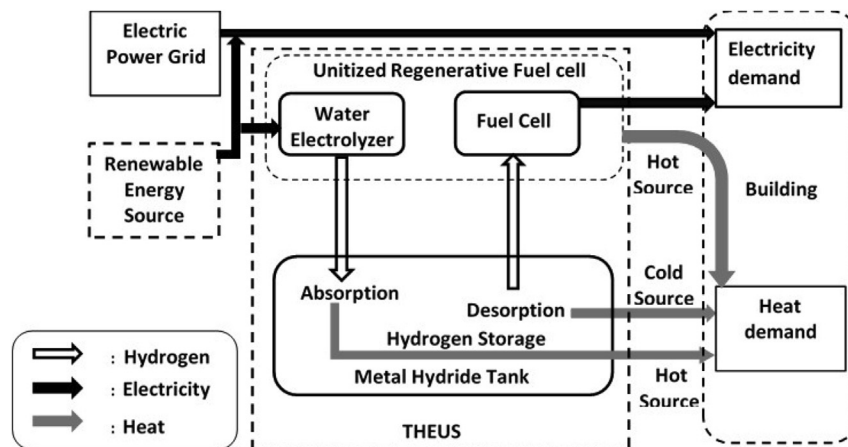


Fig. 1 – Concept of Totalized Hydrogen Energy Utilization System (THEUS).

was $\pm 1.5\%$ (FS). The URFC in WE mode was capable of producing hydrogen of about 2.3 kg/day with the maximum DC stack power of 4.4 kW. Maximum hydrogen flow rate was 17.8 NL/min and maximum output pressure was 0.97 MPa (Gauge) to avoid the High Pressure Gas Safety Act in Japan. Technical specifications of the URFC are listed in Table 1. Hydrogen produced by the URFC was stored in the MHT. Molecular sieve was used for removing the water vapor in hydrogen gas during WE mode operation. The URFC in FC mode was capable of producing 0.68 kW DC stack power. Average hydrogen supply rate to the FC was 7 NL/min at 0.68 kW stack power and the corresponding minimum supply pressure was 0.025 (Gauge) MPa. It is noted that the URFC is close to a WE rather than a FC. Input hydrogen to the FC was supplied by the MHT in desorption mode.

Fig. 3 shows the piping diagram and various components of URFC. The solid arrows and dashed arrows show the flow direction in WE mode and FC mode, respectively. As for the URFC, Nafion 115 membrane was used as the solid polymer electrolyte in the URFC stack. The URFC stack was made of ten electrochemical cells (250 cm² active area each). Platinum was used as electrocatalyst at hydrogen side and mixed catalyst of platinum and iridium black was used as electrocatalyst at oxygen side in the URFC [30]. Temperature of the URFC stack was controlled by water circulation during the operation of the URFC. Auxiliaries used in the URFC were DC power supply, electronic load, circulation pump, hydrogen circulation pump,

and air blower. Gas-liquid separation tanks were used to separate the gases (H₂ and O₂) and water. In WE mode, direct current was supplied by a DC power supply (Kikusui Electronics Co., PAT30-266T) and in FC mode, an electronic load (Kikusui Electronics Co., PLZ1004W) was applied. The URFC was connected to a thermostat bath (SMC-HRS024-A-20) for removing heat from the stack and heat exchangers in the URFC. Volumetric flow rate of the circulation fluid was 14 L/min. Circulation water inlet temperature was fixed at 12 °C. The purpose of supplying low temperature circulation water at 12 °C was to condense the water vapor in hydrogen gas during WE mode operation and also to recover the heat from all the heat exchangers in the URFC. In FC operation, unused hydrogen from the reaction was returned to the cell stack by using a circulation pump to improve the fuel consumption rate.

Fig. 4 shows the schematic of the cylindrical MHT for the THEUS. The tank body was made of stainless steel. The inner diameter was 155.2 mm, the outer diameter was 165.2 mm, and the total length was 1 m. The MHT was filled with 50 kg of MmNi₅ (AB₅) type metal hydride alloy and the total weight of the tank was approximately 85 kg. A double coil type heat exchanger, made of copper, was used to recover the reaction heat of the metal hydride during absorption/desorption process. A sheathed thermocouple with three measurement points was inserted in the tank. It was fixed at 55 mm away from the center of the side wall and the measurement points

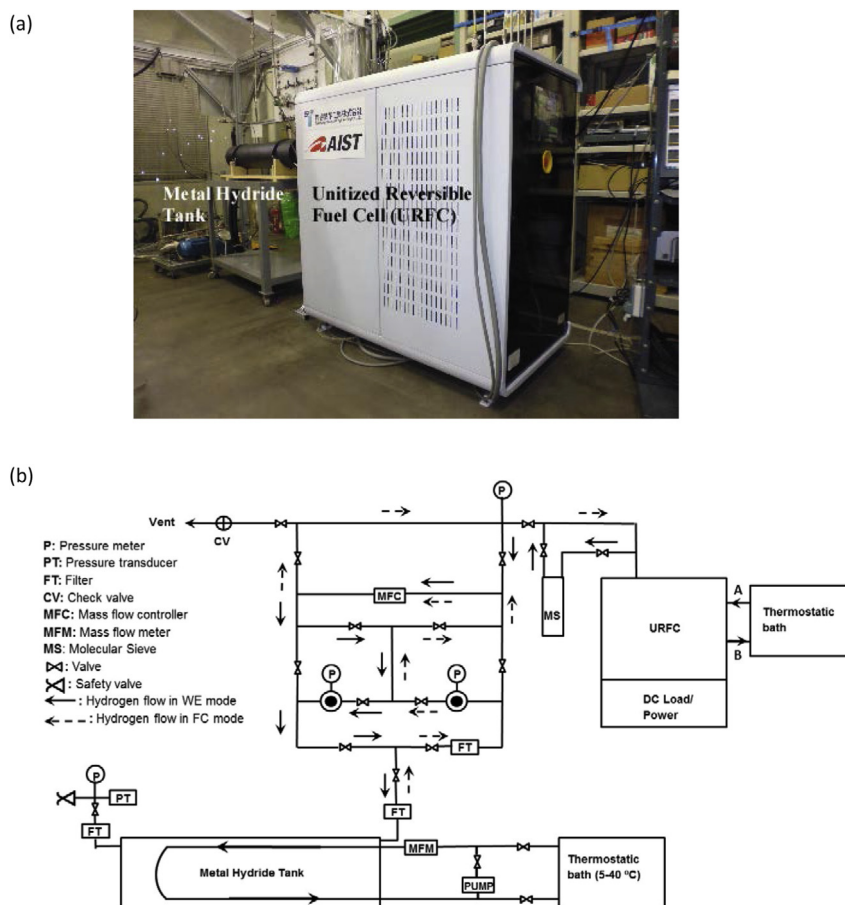


Fig. 2 – (a) Picture of the lab. scale THEUS developed at AIST. (b) Schematic of the lab. scale THEUS experimental set-up.

were 25 cm, 15 cm, and 5 cm from the outer surface of the side wall (TC-O25, TC-O15, and TC-O5), respectively. One more sheathed thermocouple with one measurement point was set at 26 mm from the center of the side wall and 10 cm from the outer surface of the side wall (TC-C10). A pressure transducer (Kyowa PGS-50KA) was used to measure the pressure in the MHT. A double screen stainless steel mesh was inserted along the inner surface of the MHT to enhance the hydrogen injection and extraction. The double coil type heat exchanger was installed in the MHT and which was connected to a thermostatic bath (SMC-HRS024-W-20) for supplying the circulating water at a desired temperature. Volumetric flow rate of the circulation water was measured by a flow meter (Keyence FD-SS2A) with an accuracy of $\pm 4\%$. Design details of the MHT used in the present experimental study was written in an earlier publication by the authors [31].

Performance parameters

Hydrogen energy output from the URFC in WE mode is calculated by using the following formula:

$$Q_{WE_OUT} = V_{H_2} \times HHV \quad (1)$$

where V_{H_2} is volume of hydrogen absorbed/desorbed by the metal hydride and HHV (J/m^3) is the higher heating value of the hydrogen. The hydrogen energy input in FC mode (Q_{FC_IN}) is also defined in a similar way.

Heat output from the URFC to thermostatic bath in WE mode and FC mode is calculated from the following equation:

$$Q_{WE_HEAT} \text{ or } Q_{FC_HEAT} = C_{pcf} \rho_{cf} \Delta T_{cf} \int \dot{V}_{cf} dt \quad (2)$$

where C_p is specific heat of the circulation fluid at constant pressure, ρ is the density, ΔT is the temperature difference between the circulation fluid outlet and inlet, and $\dot{V}$ is the volumetric flow rate. The subscript *cf* indicates the circulation fluid.

Total energy input and output of the WE are defined as:

$$Q_{WE_TI} = (Q_{WE_IN} + Q_{WE_ACCS}) \quad (3)$$

$$Q_{WE_TO} = (Q_{WE_OUT} + Q_{WE_HEAT}) \quad (4)$$

where Q_{WE_IN} and Q_{WE_ACCS} are power input to the WE and accessories (cooling pump and water pump), respectively.

Efficiency of the WE can be expressed as:

$$\eta_{WE} = Q_{WE_OUT} / Q_{WE_TI} \times 100 \quad (5)$$

Efficiency of the WE including heat output from the URFC is defined as:

$$\eta_{WE+H} = Q_{WE_TO} / Q_{WE_TI} \times 100 \quad (6)$$

Efficiency of the WE including heat output from the URFC and metal hydride heat recovery output is defined as:

$$\eta_{WE+H+MH} = (Q_{WE_TO} + Q_{MH_ABS}) / Q_{WE_TI} \times 100 \quad (7)$$

Total energy input and output of the FC are defined as:

$$Q_{FC_TI} = Q_{FC_IN} \quad (8)$$

$$Q_{FC_TO} = (Q_{FC_HEAT} + Q_{FC_OUT} - Q_{FC_ACCS}) \quad (9)$$

where Q_{FC_OUT} and Q_{FC_ACCS} are power output from the FC and power input to accessories (cooling pump, water pump and air blower), respectively.

Efficiency of the FC can be expressed as:

$$\eta_{FC} = (Q_{FC_OUT} - Q_{FC_ACCS}) / Q_{FC_TI} \times 100 \quad (10)$$

Efficiency of the FC including heat output from the URFC is defined as:

$$\eta_{FC+H} = Q_{FC_TO} / Q_{FC_TI} \times 100 \quad (11)$$

Efficiency of the FC including heat output from the URFC and metal hydride heat recovery output is defined as:

$$\eta_{FC+H+MH} = (Q_{FC_TO} + Q_{MH_DES}) / Q_{FC_TI} \times 100 \quad (12)$$

As mentioned earlier, desorption process is endothermic process, but in the THEUS, the endothermic heat was treated as a recovered cold energy output. Therefore, the recovered cold energy output (Q_{MH_DES}) was added to the Q_{FC_TO} in Eq. (12).

Total energy output and input of the URFC are defined as:

$$Q_{TO} = (Q_{WE_HEAT} + Q_{FC_HEAT} + Q_{FC_OUT} - Q_{FC_ACCS}) \quad (13)$$

$$Q_{TI} = (Q_{WE_IN} + Q_{WE_ACCS}) \quad (14)$$

Total efficiency of the URFC can be expressed as:

$$\eta_{URFC} = Q_{TO} / Q_{TI} \times 100 \quad (15)$$

In this study, total efficiency of the THEUS is defined as

$$\eta_{THEUS} = (Q_{TO} + Q_{MH_ABS} + Q_{MH_DES}) / Q_{TI} \times 100 \quad (16)$$

In Eq. (16) also Q_{MH_DES} is treated as the energy output. The uncertainty of the total efficiency was $\pm (2.7\% \leq \text{uncertainty of } \eta_{THEUS} \leq 4.4\%)$. The uncertainty was obtained from the uncertainties of power output, heat output from the URFC and recovered reaction heat. The uncertainty of the heat output and the recovered reaction heat was about $\pm 4.4\%$. But the accuracies of the measurement equipment's for the voltage and current are higher than the accuracies of mass flow meters. Therefore, the minimum uncertainty was estimated about $\pm 2.7\%$ in this case.

The recovered reaction heat, $Q_{MH_ABS/DES}$ is calculated by using the following equation:

$$Q_{MH_ABS/DES} = C_{pcw} \rho_{cw} \left| \Delta T_{cw} \right| \int \dot{V}_{cw} dt \quad (17)$$

The subscript *cw* indicates the circulation water.

The metal hydride reaction heat recovery rate, ϵ can be defined as:

$$\epsilon = Q_{MH_ABS/DES} / Q_{MH_REACT} \times 100 \quad (18)$$

where Q_{MH_REACT} is the reaction heat of the metal hydride. The value of Q_{MH_REACT} is estimated by $Q_{MH_REACT} = V_{H_2} \times \Delta H$, where V_{H_2} is the volume of hydrogen absorbed/desorbed by the metal hydride, and ΔH is the reaction heat per volume of the metal hydride. The uncertainty in the estimation of the reaction heat recovery rate was about $\pm 5\%$ and it depends on the

Table 1 – Technical specifications of the URFC.

Cell number	10		
Cell surface	250 cm ²		
Membrane	Nafion 115		
	WE mode		FC mode
Maximum DC stack power	4.4 kW		0.68 kW
Maximum flow rate	17.8 NL/min		7 NL/min
Max/min pressure	0.97 MPa (Gauge)		0.025 MPa (Gauge)
Start-up time	8 min		8 min

accuracy of the mass flow meter used for measuring the circulation water and hydrogen [31].

Results and discussion

In the following sections, the results obtained from the experimental studies are presented. The performance of the lab. scale THEUS was investigated. The energy efficiency of the individual component characterizations and total system behavior are reported and discussed.

Individual WE mode and FC mode tests results

The THEUS was tested at full power in WE mode. The power input to the URFC in WE mode was fixed at 4.4 kW (250 A, maximum current). Fig. 5(a) shows the power and hydrogen flow rate. In Fig. 5(a), negative power indicates the power input to the URFC and positive flow rate indicates hydrogen production from the URFC. The URFC was operated for 4 h 32 min and after that it was stopped due to the maximum pressure limit of the URFC. Hydrogen flow rate was in the range of 17.5–18.4 NL/min with the average flow rate of 17.8 NL/min. Hydrogen flow data is listed in Table 2. The total amount of hydrogen absorbed in the MHT was 4870 NL. The spikes observed in the hydrogen production rate are might be due to pressure regulation of the output hydrogen from the URFC. Fig. 5(b) shows the heat output from the URFC during the full power WE mode operation. The circulation fluid carries the heat generated from the URFC to thermostatic bath. The spikes appeared due to the heat output from the heat exchanger which was connected to the cell stack. The circulation pump for the cell stack was intermittently operated to control the temperature of the cell stack. The heat output (Q_{WE_HEAT}) was calculated using Eq. (2), which was 6.56 MJ and the average heat output rate from the URFC was 0.4 kW.

The dynamic Pressure-Composition-Temperature (P-C-T) isotherm, temperatures profiles measured inside the tank, inlet and outlet temperature of the circulation water were measured. Fig. 6(a) shows the dynamic P-C-T isotherm for WE (absorption) operation. At the starting point, the selected initial composition H/M of the MHT was 0.18. In this test, the circulation water flow rate was fixed at 1.83 L/min to maintain 3 °C temperature difference between the inlet and outlet temperature of the circulation water. The circulation water inlet temperature was fixed at 32 °C. After absorption process, the final composition H/M of the MHT was 0.80. Fig. 6(b) shows the variation of temperatures inside the tank, inlet, and outlet temperature of the circulation water. It can be seen from Fig.

6(b) that the temperatures inside the tank were increased steeply in the first 20 min and after that they were gradually increased. On the other hand, the circulation water outlet temperature was remained steady about 35 min after the absorption process started. The recovered reaction heat from the MHT was 5.16 MJ (Q_{MH_ABS}). In this test, the metal hydride reaction heat recovery rate was 82%. For operating the WE about 9 h operation in continuous 24 h test, it is confirmed that two MHT's are needed. Hence, the input power was reduced to test the THEUS in continuous 24 h cycle operation, which was discussed in the Continuous operation results section.

For preparation of the continuous 24 h cycle operation, the power input to the THEUS in WE mode was fixed at 2.05 kW Fig. 7(a) shows the power and hydrogen flow rate during WE operation and Fig. 7(b) shows the power and hydrogen flow rate during full power FC operation. In Fig. 7(b), the positive power indicates the power output from the URFC and negative hydrogen flow rate indicates hydrogen supplied to the URFC from the MHT. Hydrogen flow rate was in the range of 7.8–8.8 NL/min with the average flow rate of 8.5 NL/min. The total amount of hydrogen absorbed in the MHT during WE operation was 5506 NL. The FC was operated for 13 h and the hydrogen flow rate was in the range of 3–23 NL/min with the average flow rate of 7 NL/min as shown in Table 2. The stack power of the FC was increased step by step until it reached to 0.68 kW. The FC power output was increased with increase in stack temperature. The main reason for this protocol is to avoid flooding in the cell stack. In Fig. 7(b), the spikes can be seen in the hydrogen flow rate. They were appeared due to the hydrogen recirculation for every 5 min by the hydrogen circulation pump. The main role of the pump is improving the fuel consumption as mentioned before, and in addition, it prevent the flooding in the cell stack. The hydrogen flow rate also increased proportional to the FC power output. The total amount of hydrogen desorbed from the MHT during FC operation was 5445 NL. Fig. 8(a) and Fig. 8(b) show the heat output from the URFC during WE mode and FC mode operation, respectively. From these figures it is noted that the heat output from the URFC in FC mode is higher than the heat output from the URFC in WE mode. The heat output from the URFC in WE mode and FC mode were 9.59 MJ and 19.47 MJ, respectively. The average heat output rates from the URFC in WE mode and FC mode were 0.25 kW and 0.42 kW, respectively. It is noted that the spikes in the FC mode are also longer than those in the WE mode.

Fig. 9 show the results of the MHT. Fig. 9(a) shows the dynamic P-C-T isotherm for both WE (absorption) and FC (desorption) operation. At the starting point, the selected initial composition H/M of the MHT was 0.18. In WE mode

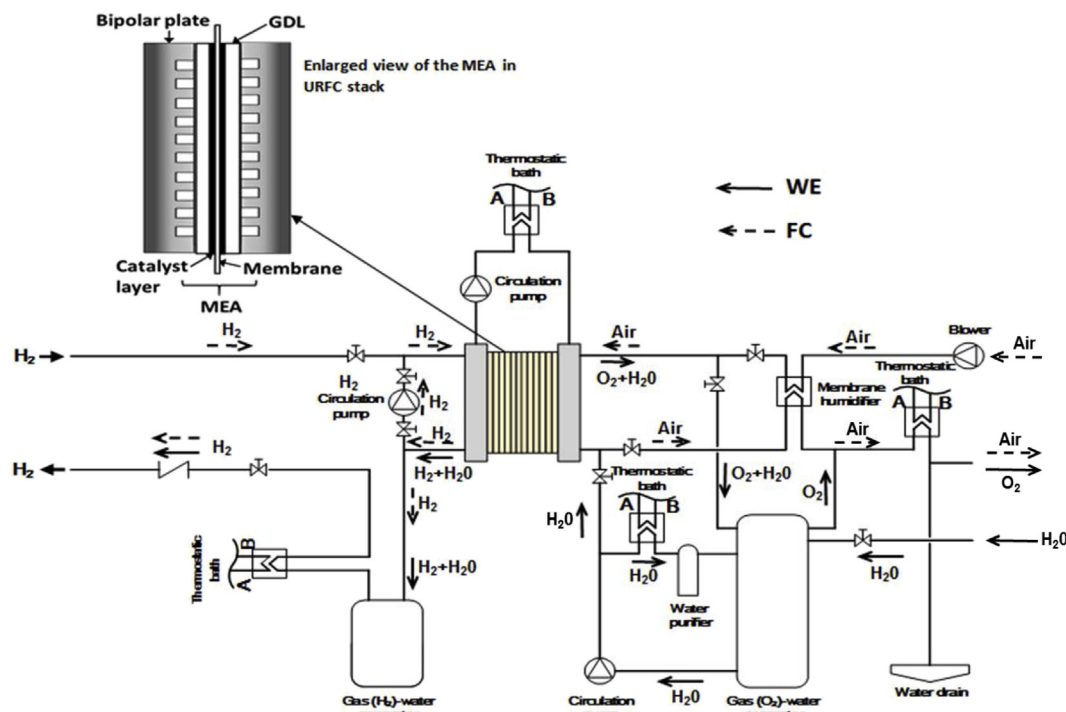


Fig. 3 – Flow field and various components of the URFC.

operation the circulation water flow rate was fixed at 0.77 L/min. The circulation water inlet temperature was fixed at 32 °C and 12 °C in WE mode operation and FC mode operation, respectively. After absorption process, the final composition H/M of the MHT was 0.88 and at the end of the desorption process, it was 0.18. The final pressure of the MHT during WE

and FC mode operation were 0.93 MPa (0.92 MPa Gauge) and 0.139 MPa, respectively. The pressure range was kept according to the specification of URFC. Table 3 shows the operating conditions of the MHT. Fig. 9(b) shows the variation of temperatures inside the tank, inlet and outlet temperatures of the circulation water during WE operation. Temperatures inside

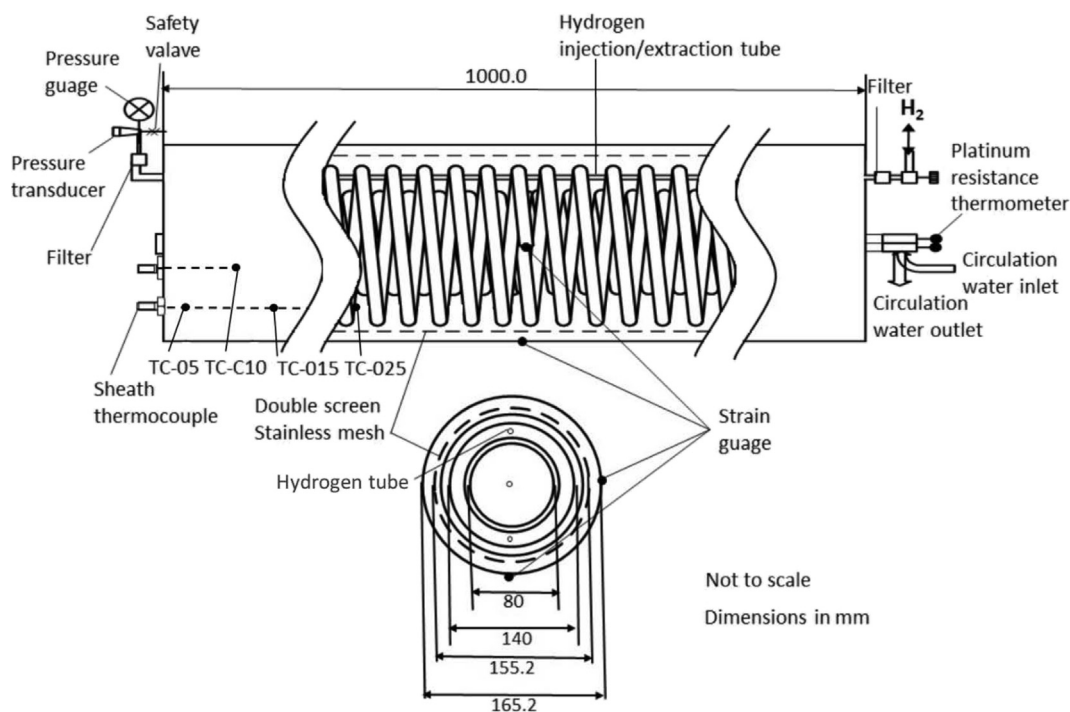


Fig. 4 – Schematic of the MHT.

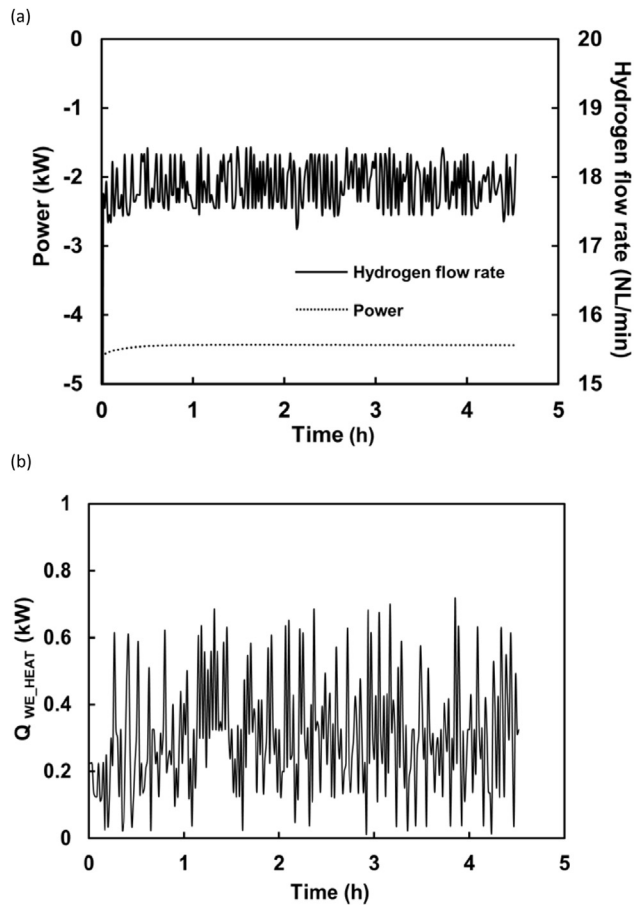


Fig. 5 – Results for the full power WE mode operation. (a) Power and hydrogen flow rate. (b) Heat output from the URFC.

the tank in WE operation were relatively uniform and steady except the TC-O5 temperature profile. This was due to the position of the thermocouple TC-O5 was far from the heat exchanger. It is noted that the temperature of the MHT is lower than the temperature shown in Fig. 6(b). Therefore, the pressure in MHT was decreased, so the composition became large. The circulation water outlet temperature reached to 35 °C roughly about 1 h after the absorption started and after that it was remained steady. Fig. 9(c) shows that the temperatures inside the tank, inlet and outlet temperatures of the

Table 2 – Hydrogen flow data of the THEUS experiments.

URFC		Amount of H ₂ absorption/desorption (NL)	Average flow rate (NL/min)
Full power	WE	4870	17.8
Separated	WE	5506	8.5
	FC	5445	7.0
Day1	WE	5398	8.4
	FC	5476	7.0
Day2	WE	5397	8.4
	FC	5484	7.0
Day3	WE	5351	8.4
	FC	5470	7.0

circulation water during FC operation. The temperatures inside the tank were relatively uniform and decreased continuously. In FC mode operation the circulation water flow rate was fixed at 0.39 L/min to maintain 5 °C temperature difference between the inlet and outlet temperature of the circulation water. The circulation water outlet temperature slowly decreased to 7 °C because the hydrogen consumption of the FC was increased step by step as shown in Fig. 7(b). The recovered reaction heat from the MHT was 5.61 MJ (Q_{MH_ABS}) during WE mode operation and 5.40 MJ (Q_{MH_DES}) during FC mode operation. The metal hydride reaction heat recovery rate was 79% and 74% during FC operation, respectively.

Continuous operation results

For more practical point of view, the THEUS was tested continuously for 3 days. THEUS was operated in WE mode operation for 10 h 44 min and in FC mode operation for 13 h. The switching time between WE and FC in URFC was about 8 min. Power input to the THEUS in WE mode and power output from the THEUS in FC mode were also fixed at 2.05 kW and 0.68 kW. Fig. 10 shows the power and hydrogen flow rate during Day1 operation. In Fig. 10, the ‘sign’ indication was

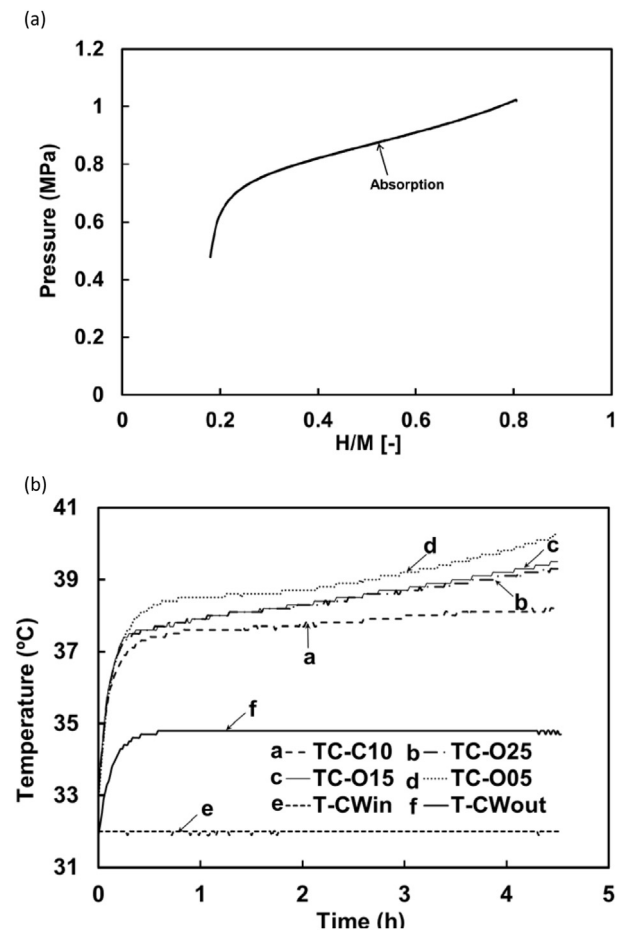


Fig. 6 – Results for the full power WE mode test of the MHT. (a) Dynamic P-C-T isotherm. (b) Temperature profiles measured inside the tank, inlet and outlet temperature of the circulation water during WE mode operation.

similar to Fig. 7. It can be observed that URFC during FC mode operation takes about 1 h 14 min to reach the predefined output and after that the power output was steady until the end of the operation. The hydrogen flow rate was in the range of 7.8–8.8 NL/min with the average flow rate of 8.4 NL/min during WE mode operation. It was in the range of 3–23 NL/min with the average flow rate of 7 NL/min during FC mode operation (see Table 2). Fig. 11(a) and Fig. 11(b) show the heat output from the URFC during WE mode and FC mode (Day2) operation respectively. The volumetric flow rate of the circulation fluid was 14 L/min and inlet temperature was fixed at 12 °C. Fig. 11(a and b) show that the heat output pattern is similar to separated test. The heat output from the URFC in WE mode and FC mode (Day2) were 8.88 MJ and 17.81 MJ, respectively. The average heat output rate from the URFC in WE mode and FC mode were 0.23 kW and 0.38 kW, respectively.

Fig. 12(a) shows the dynamic P-C-T isotherm of the metal hydride during 3 days continuous operation. During the first absorption process the P-C-T isotherm followed the separated test behavior, after that the pressure profile showed a closed loop: switching time, the end point of desorption

process, again switching time, and vice versa. In the continuous operation, the starting point of desorption process at higher temperature and absorption process at lower temperature as compared with the separated test except the first absorption process. The maximum composition H/M of the MHT was 0.87. The maximum and minimum pressure of the MHT during WE and FC mode operation were 1.02 MPa and 0.136 MPa respectively. Fig. 12(b) shows the circulation water outlet temperature during 3 days continuous operation. The Day1 absorption temperature profile was similar to separated test result shown in Fig. 9(b). For recovering heat from the MHT during the switching time, the circulation water flow rate was increased to 1.12 L/min because at the end of the absorption process, the MHT temperature was still high. After the switching, the URFC was started in FC mode, but the temperature of the MHT was still higher than the assumed desorption temperature 12 °C. Hence, the MHT was self-cooled by using the endothermic reaction heat of metal hydride during desorption process. The circulation water was recirculated using a bypass valve and pump shown in Fig. 2(b) without fed into the thermostatic bath. Once the circulation water output temperature was reached to 12 °C again the circulation water was fed into the thermostatic

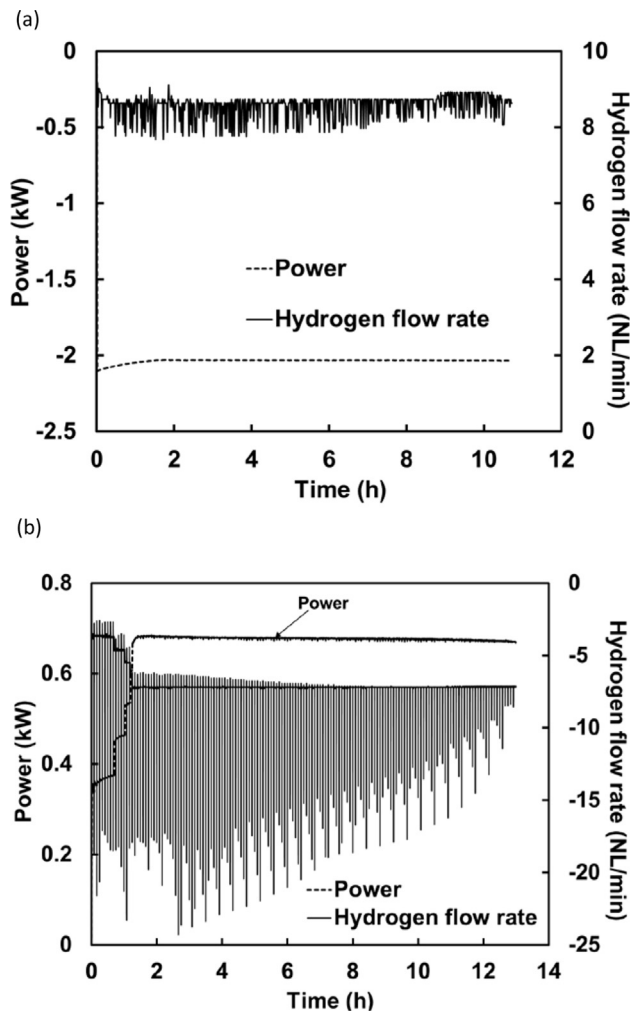


Fig. 7 – Results for the power and hydrogen flow rate. (a) WE mode operation. (b) FC mode operation.

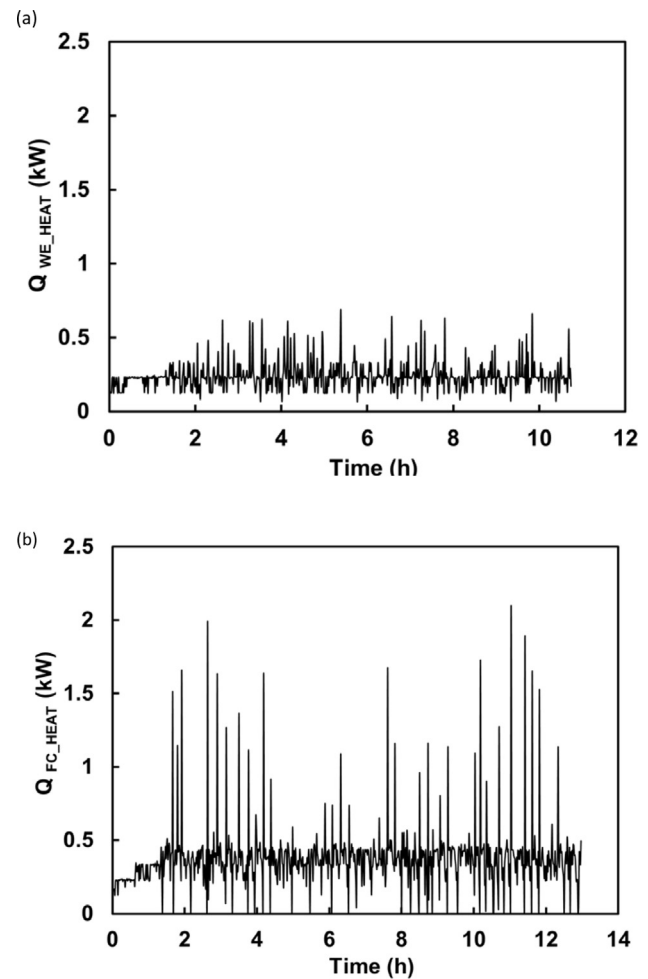


Fig. 8 – Heat output from the URFC. (a) WE mode operation. (b) FC mode operation.

bath and the circulation water flow rate was decreased to 0.39 L/min. At the end of the desorption process, the temperature of the metal hydride was still below 12 °C. For recovering the cold energy from the MHT during the switching time, the circulation water flow rate was increased

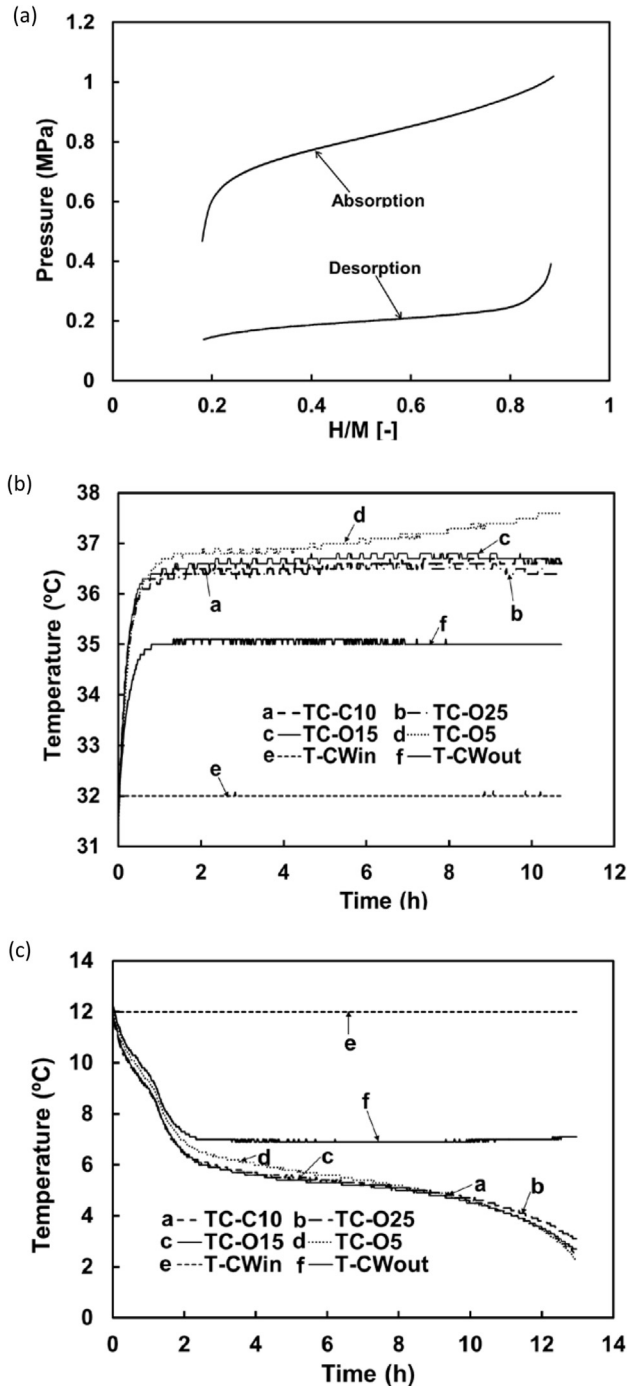


Fig. 9 – Results for the separated test of the MHT. (a) Dynamic P-C-T isotherm. (b) Temperature profiles measured inside the tank, inlet and outlet temperature of the circulation water during WE mode operation. (c) Temperature profiles measured inside the tank, inlet and outlet temperature of the circulation water during FC mode operation (locations are illustrated in Fig. 4).

Table 3 – Operating conditions of the MHT.

Mode of operation	WE	FC
Time	10 h: 44 min	13 h
Pressure	$P < 0.97$ MPa (G)	$P > 0.025$ MPa (G)
Circulation water inlet temperature	32 °C	12 °C
Average H ₂ flow rate (NL/min)	8.5	7.0
Circulation water flow rate (l/min)	0.77	0.39

again to 1.12 L/min. After switching, the URFC was started in WE mode, but the temperature of the MHT was still lower than the assumed absorption temperature of 32 °C. Again the MHT was self-heated by using exothermic reaction heat of metal hydride during absorption process. The flow direction of the circulation water was recirculated by using bypass valve and pump without fed into the thermostatic bath. Once the circulation water output temperature was reached to 32 °C again the circulation water was fed into the thermostatic bath and the circulation water flow rate was decreased to 0.77 l/min. The above procedure was repeated for 3 days continuously. The variation in circulation water outlet temperature after the first absorption process was similar in 3 days. In these experiments, the reaction heat recovery rates were 81% (Day1), 67% (Day2), 67% (Day3) in absorption processes and 62% in all desorption processes. The reaction heat recovery rates were smaller than the separated test results. This was due to the fact that some part of the reaction heat was used to self-cooling or self-heating of the MHT during desorption or absorption process. The reaction heat recovery rates were 10% smaller than the unit testing for the MHT [31] due to short switching time (8 min) where as in that case the interval time was 1 h Fig. 12(c) show the temperature profiles measured inside the tank during Day1 and the WE operation in Day2. It can be seen that the starting period of Day2 temperature profiles were different from the temperature profiles of Day1. This is due to the fact that in Day1 experiment, the MHT initial temperature was at 32 °C and in Day2 initial temperature was depending on the previous FC operation. The MHT tank takes some time to reach 32 °C by using

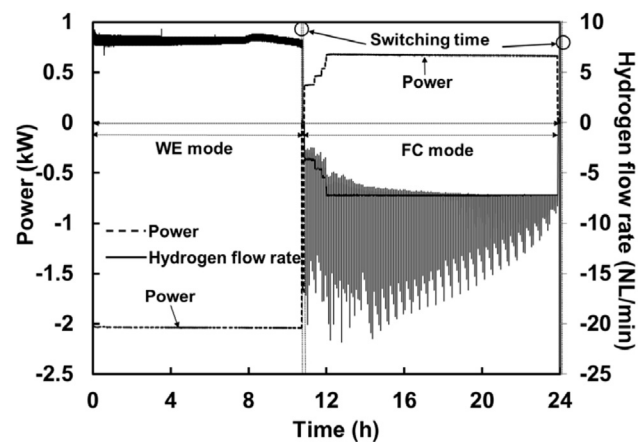


Fig. 10 – Power and hydrogen flow rate during Day1 operation.

exothermic reaction heat. The Day3 temperature profiles were the same as the profiles of Day2, so they were omitted. THEUS was successfully tested continuously for 3 days without any interruption. Hence, this study confirms that the stability of THEUS upon cycling.

Energy efficiency evaluation of the THEUS

Tables 4 and 5 shows the performance data of the THEUS in the full power WE mode test and the separated test discussed in the Individual WE mode and FC mode tests results section. The total energy supplied to the auxiliaries and total power input (including accessories) to the URFC during the full power WE operation were 2.30 MJ and 75.19 MJ. The heat output from the URFC was 6.56 MJ and the total energy output from the URFC was 68.84 MJ. The WE efficiency including heat output from the URFC and heat output from the MHT was 98%.

In the separated test, the THEUS was operated for 10 h 44 min in WE mode and 13 h in FC mode. The total energy supplied to the auxiliaries during the WE operation was 5.41 MJ and it was 13.66 MJ during the FC operation. The heat output from the URFC in WE mode and FC mode were 9.59 MJ and 19.47 MJ, respectively. The total power input (including

accessories) to the URFC in WE mode operation was 84.10 MJ and the total net power output ($Q_{FC_OUT} - Q_{FC_ACCS}$) from the URFC in FC mode operation was 16.90 MJ. The recovered thermal energy from the MHT was 5.61 MJ in WE operation and 5.40 MJ in FC operation. The efficiencies of the URFC in WE mode and FC mode were 84% and 24%, respectively. The WE

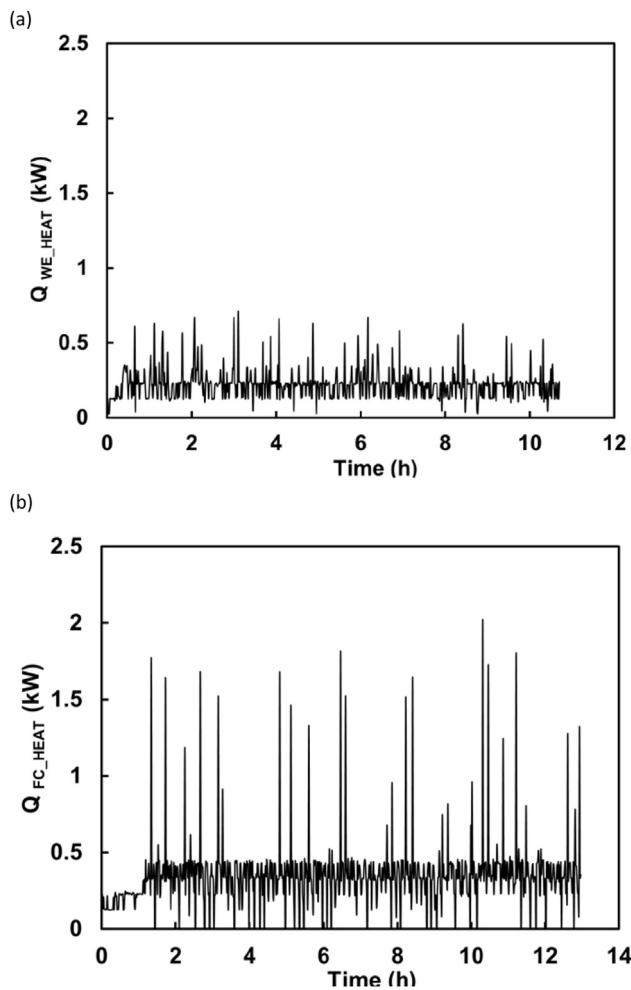


Fig. 11 – Heat output from the URFC during Day2 operation. (a) WE mode operation. (b) FC mode operation.

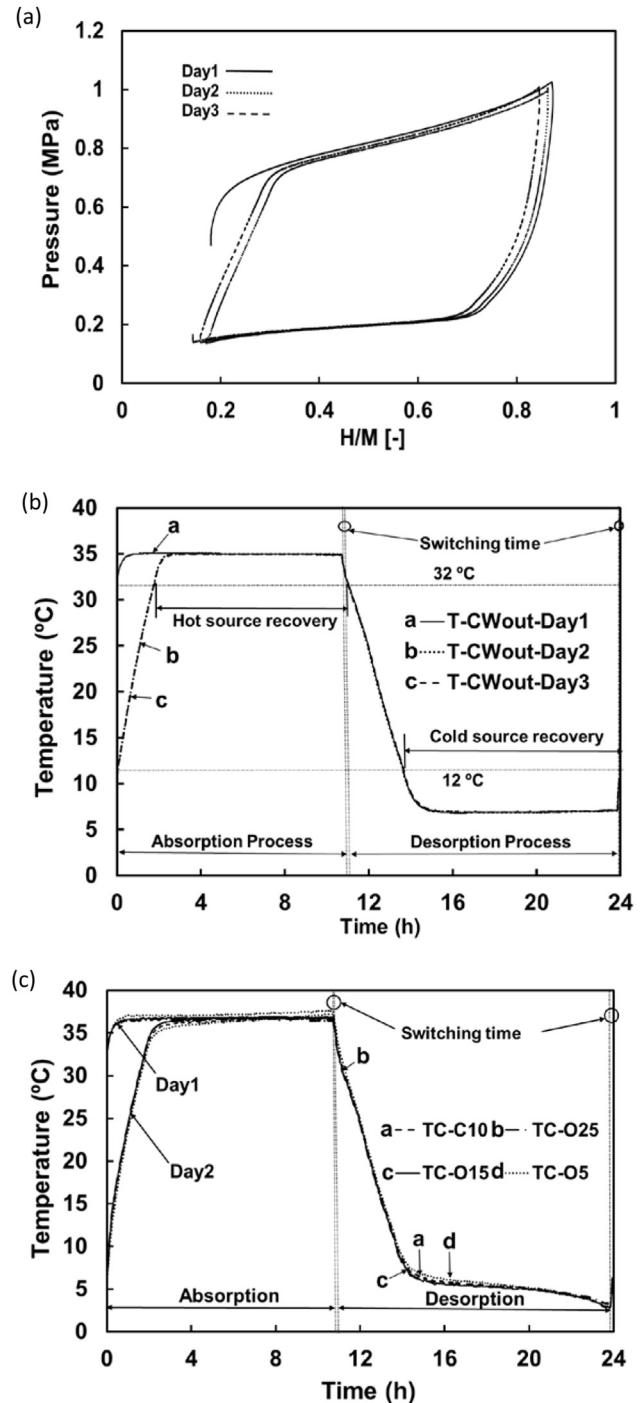


Fig. 12 – Results for the continuous three day test of the MHT. (a) Dynamic P-C-T isotherm. (b) Outlet temperature of the circulation water. (c) Temperature profiles measured inside the tank during Day1 and WE mode operation in Day2.

Table 4 – Performance data of the THEUS for full power WE mode operation.

		Output (MJ)		Input (MJ)		MH output (MJ)
WE	Q_{WE_OUT}	62.28	Q_{WE_IN}	72.89	Q_{MH_ABS}	5.16
	Q_{WE_HEAT}	6.56	Q_{WE_ACCS}	2.30		
	Q_{WE_TO}	68.84	Q_{WE_TI}	75.19	ϵ_{ABS}	82%
	η_{WE}	83%	η_{WE+H}	92%	$\eta_{WE+H+MH}$	98%

Table 5 – Performance data of the THEUS for the individual WE mode and FC mode tests.

		Output (MJ)		Input (MJ)		MH output (MJ)
WE	Q_{WE_OUT}	70.42	Q_{WE_IN}	78.69	Q_{MH_ABS}	5.61
	Q_{WE_HEAT}	9.59	Q_{WE_ACCS}	5.41		
	Q_{WE_TO}	80.01	Q_{WE_TI}	84.10	ϵ_{ABS}	79%
	η_{WE}	84%	η_{WE+H}	95%	$\eta_{WE+H+MH}$	102%
FC	Q_{FC_OUT}	30.56	Q_{FC_IN}	69.64	Q_{MH_DES}	5.40
	Q_{FC_HEAT}	19.47	Q_{FC_ACCS}	13.66		
	Q_{FC_TO}	36.37	Q_{FC_TI}	69.64	ϵ_{DES}	74%
	η_{FC}	24%	η_{FC+H}	52%	$\eta_{FC+H+MH}$	60%
	Q_{TO}	45.96	Q_{TI}	84.10		
			η_{URFC}	55%	η_{THEUS}	68%

efficiency was almost same as the efficiency of the full power operation although it was operated at a half of input power. The efficiency of URFC in WE mode operation was high because there was no power input for the dehumidification. On the other hand, the efficiency of the URFC in FC mode was low as compared to the WE mode operation. Due to the over potential difference between the WE and FC, the efficiency of

FC is lower than the WE. The $\eta_{WE+H+MH}$ is 2% higher than 100% but this is within the uncertainty range. Total efficiency of the URFC was 55% and total efficiency of the THEUS was 68%, respectively.

Table 6 shows the performance data of the THEUS in a continuous 3 day operation. Power input to the URFC in WE mode and power output from the URFC in FC mode were same

Table 6 – Performance data of the THEUS in continuous 3 day operation.

		Output (MJ)		Input (MJ)		MH output (MJ)	
Day1	WE	Q _{WE-OUT}	69.04	Q _{WE-IN}	78.79	Q _{MH-ABS}	5.65
		Q _{WE-HEAT}	8.95	Q _{WE-ACCS}	5.41		
		Q _{WE-TO}	77.99	Q _{WE-TI}	84.20	ε _{ABS}	81%
		η _{WE}	82%	η _{WE+H}	93%	η _{WE+H+MH}	99%
	FC	Q _{FC-OUT}	30.46	Q _{FC-IN}	70.03	Q _{MH-DES}	4.51
		Q _{FC-HEAT}	17.62	Q _{FC-ACCS}	13.66		
		Q _{FC-TO}	34.42	Q _{FC-TI}	70.03	ε _{DES}	62%
		η _{FC}	24%	η _{FC+H}	49%	η _{FC+H+MH}	56%
		Q _{TO}	43.37	Q _{TI}	84.20		
				η _{URFC}	52%	η _{THEUS}	64%
Day2	WE	Q _{WE-OUT}	69.02	Q _{WE-IN}	78.64	Q _{MH-ABS}	4.68
		Q _{WE-HEAT}	8.88	Q _{WE-ACCS}	5.41		
		Q _{WE-TO}	77.90	Q _{WE-TI}	84.05	ε _{ABS}	67%
		η _{WE}	82%	η _{WE+H}	93%	η _{WE+H+MH}	98%
	FC	Q _{FC-OUT}	30.50	Q _{FC-IN}	70.14	Q _{MH-DES}	4.51
		Q _{FC-HEAT}	17.81	Q _{FC-ACCS}	13.66		
		Q _{FC-TO}	34.65	Q _{FC-TI}	70.14	ε _{DES}	62%
		η _{FC}	24%	η _{FC+H}	49%	η _{FC+H+MH}	56%
		Q _{TO}	43.53	Q _{TI}	84.06		
				η _{URFC}	52%	η _{THEUS}	63%
Day3	WE	Q _{WE-OUT}	68.43	Q _{WE-IN}	78.54	Q _{MH-ABS}	4.64
		Q _{WE-HEAT}	8.93	Q _{WE-ACCS}	5.41		
		Q _{WE-TO}	77.36	Q _{WE-TI}	83.95	ε _{ABS}	67%
		η _{WE}	82%	η _{WE+H}	92%	η _{WE+H+MH}	98%
	FC	Q _{FC-OUT}	30.59	Q _{FC-IN}	69.96	Q _{MH-DES}	4.50
		Q _{FC-HEAT}	17.31	Q _{FC-ACCS}	13.66		
		Q _{FC-TO}	34.24	Q _{FC-TI}	69.96	ε _{DES}	62%
		η _{FC}	24%	η _{FC+H}	49%	η _{FC+H+MH}	55%
		Q _{TO}	43.17	Q _{TI}	83.95		
				η _{URFC}	51%	η _{THEUS}	62%

as the separated test. Total efficiency of the URFC in Day1 was 52% and total efficiency of the THEUS was 64%. Total efficiency of the THEUS in Day1 test was 4% lower as compared to the separated test. This is due to the reason that some part of reaction heat of metal hydride was lost in self-cooling of the MHT during FC operation. Total efficiency of the THEUS in Day2 and Day3 tests was almost same and they were slightly lower than the efficiency in Day1. This is due to the fact that some part of the reaction heat of metal hydride was lost in self-heating of the MHT during WE operation. The WE efficiency, FC efficiency and total efficiency of the URFC in Day2 were not different from those in Day3. Total efficiency of the URFC in Day2 test was 52% and Day3 test was 51%. Total efficiency of the THEUS in Day2 test was 63% and Day3 test was 62%. It was confirmed that the contribution of the MHT's reaction heat for the total efficiency was 11%.

Conclusions

Experiments were carried out on the lab. scale THEUS. Energy efficiency of the THEUS was evaluated for the full power WE mode, individual WE mode and FC mode tests, and the continuous 3 day operation test. Energy efficiency of the individual component characterizations and the total system behavior were reported in tables. Total efficiency of URFC was 55% and total efficiency of the THEUS was 68% in the case of the individual WE mode and FC mode tests. Total efficiency of the URFC was 52% and total efficiency of the THEUS was 63% in the case of the continuous WE mode-FC mode operation. The metal hydride reaction heat recovery rate in separated test was 79% during WE operation and 74% during FC operation. In the continuous test, the MH reaction heat recovery rate was 67% in the WE mode operation and 62% in the FC mode operation. The maximum and minimum pressure of the MHT during WE and FC mode operations were 1.02 MPa and 0.136 MPa, respectively. All operations for the THEUS were successfully carried out under 1.1 MPa (1.0 MPa Gauge). The total efficiency of the THEUS can be further improved by proper selection of auxiliary's power capacity, piping and flow field of the URFC.

Acknowledgments

This study was supported from the New Energy and Industrial Technology Development Organization (NEDO), the contract administration number 14103078-0. We wish to acknowledge the contributions of Dr. T. Motyka, Dr. R. Zidan, Dr. C. Corgnale at the Savannah River National Laboratory and Dr. S. Greenway at the Greenway Energy LLC in U. S. on the MHT design. We also wish to thank Prof. B. C. Hauback, Dr. O. Ulleberg, and Dr. P. Vie at the IFE in Norway, Dr. M. Dornheim, Dr. K. Taube, and Dr. J. M. Bellosta von Colbe at the HZG in Germany, Prof. A. Zuttel at the EPFL in Switzerland and Prof. S. Orimo at the Tohoku University, Japan for fruitful suggestions and discussions. We wish to thank Mr. N. Miyazaki and Mrs. S. Takagi for very helpful assistance.

REFERENCES

- [1] Penner SS. Steps towards the hydrogen economy. *Int J Hydrogen Energy* 2006;31:33–43.
- [2] Gray EM, Webb CJ, Andrews J, Shabani B, Tsai PJ, Chan SLI. Hydrogen storage for off-grid power supply. *Int J Hydrogen Energy* 2011;36:654–63.
- [3] Ipsakis D, Voutetakis S, Seferlis P, Stergiopoulos F, Elmasides C. Power management strategies for a stand-alone power system using renewable energy sources and hydrogen storage. *Int J Hydrogen Energy* 2009;34:7081–95.
- [4] Paul B, Andrews J. Optimal coupling of PV arrays to PEM electrolyzers in solar-hydrogen systems for remote area power supply. *Int J Hydrogen Energy* 2008;33:490–8.
- [5] Chávez-Ramírez AU, Cruz JC, Espinosa-Lumbreras R, Ledesma-García J, Durón-Torres SM, Arriaga LG. Design and set up of a hybrid power system (PVWT-URFC) for a stand-alone application in Mexico. *Int J Hydrogen Energy* 2013;38:12623–33.
- [6] Rekioua D, Bensmail S, Bettar N. Development of hybrid photovoltaic-fuel cell system for stand-alone application. *Int J Hydrogen Energy* 2014;39:1604–11.
- [7] Giatrakos GP, Tsoutsos TD, Mouchtaropoulos PG, Naxakis GD, Stavrakakis G. Sustainable energy planning based on a stand-alone hybrid renewable energy/hydrogen power system: application in Karpachos island, Greece. *Renew Energy* 2009;34:2562–70.
- [8] Kroniger D, Madlener R. Hydrogen storage for wind parks: a real options evaluation for an optimal investment in more flexibility. *Appl Energy* 2014;136:931–46.
- [9] MacDonald BD, Rowe AM. A thermally coupled metal hydride hydrogen storage and fuel cell system. *J Power Sources* 2006;161:346–55.
- [10] Adam A, Fraga ES, Brett DJL. Options for residential building services design using fuel cell based micro-CHP and the potential for heat integration. *Appl Energy* 2015; 138:685–94.
- [11] Balta-Ozkan N, Baldwin E. Spatial development of hydrogen economy in a low-carbon UK energy system. *Int J Hydrogen Energy* 2013;38:1209–24.
- [12] Maclay JD, Brouwer J, Scott Samuelsen G. Dynamic analyses of regenerative fuel cell power for potential use in renewable residential applications. *Int J Hydrogen Energy* 2006;31:994–1009.
- [13] Keliang Z, Ferreira JA, De Haan SWH. Optimal energy management strategy and system sizing method for standalone photovoltaic-hydrogen systems. *Int J Hydrogen Energy* 2008;33(2):477–89.
- [14] Genc G, Celik M, Serdar Genç M. Cost analysis of wind electrolyzer-fuel cell system for energy demand in Pınarbası-Kayseri. *Int J Hydrogen Energy* 2012;37:12158–66.
- [15] Gahleitner G. Hydrogen from renewable electricity: an international review of power-to-gas pilot plants for stationary applications. *Int J Hydrogen Energy* 2013;2039–61.
- [16] Valverde L, Rosa F, Bordons C. Design, planning and management of a hydrogen-based microgrid. In: *Industrial Informatics, IEEE Transactions*, 9; 2013. p. 1398–404.
- [17] Valverde L, Rosa F, Del Real AJ, Arce A, Bordons C. Modeling, simulation and experimental set-up of a renewable hydrogen-based domestic microgrid. *Int J Hydrogen Energy* 2013;38:11672–84.
- [18] Verma A, Basu S. Feasibility study of a simple unitized regenerative fuel cell. *J Power Sources* 2004;135:62–5.
- [19] Grigoriev SA, Millet P, Porembsky VI, Fateev VN. Development and preliminary testing of a unitized regenerative fuel cell based on PEM technology. *Int J Hydrogen Energy* 2011;36:4164–8.

- [20] Tange M, Maeda T, Nakano A, Ito H, Kawakami Y, Masuda M, et al. Experimental study of hydrogen storage with reaction heat recovery using metal hydride in a totalized hydrogen energy utilization system. *Int J Hydrogen Energy* 2011;36:11767–76.
- [21] Corgnale C, Motyka T, Greenway S, Perez-Berrios MJ, Nakano A, Ito H, et al. Metal hydride bed system model for renewable source driven regenerative fuel cell. *J Alloy Compd* 2013;580(S1):S406–9.
- [22] Nakano A, Ito H, Maeda T, Munakata T, Motyka T, Corgnale C, et al. Study on a metal hydride tank to support energy storage for renewable energy. *J Alloy Compd* 2013;580-S1(15):S418–22.
- [23] Nakano A, Maeda T, Ito H, Masuda M, Kawakami Y, Tange M, et al. Study on absorption/desorption characteristics of a metal hydride tank for boil-off gas from liquid hydrogen. *Int J Hydrogen Energy* 2012;37:5056–62.
- [24] Nakano A, Maeda T, Ito H, Masuda M, Kawakami Y, Kato A, et al. Small-scale hydrogen liquefaction with a two-stage Gifford-McMahon cycle refrigerator. *Int J Hydrogen Energy* 2010;35:9088–94.
- [25] Zuttel A. Hydrogen storage methods. *Naturwissenschaften* 2004;91:157–72.
- [26] Okada Y, Sasaki E, Watanabe E, Hyodo S, Nishijima H. Development of dehydrogenation catalyst for hydrogen generation in organic chemical hydride method. *Int J Hydrogen Energy* 2006;31:1348–56.
- [27] Lototsky MV, David MW, Tolj I, Klochko YV, Sekhar BS, Chidziva S, et al. Metal hydride systems for hydrogen storage and supply for stationary and automotive low temperature PEM fuel cell power modules. *Int J Hydrogen Energy* 2015;40:11491–7.
- [28] Tamhankar S, Boyd T, Gulamhusein A, Golben M, DaCosta D. Integrated hydrogen production, purification and compression system. DoE hydrogen program, project PD7, http://www.hydrogen.energy.gov/pdfs/review08/pd_7_tamhankar.pdf.
- [29] Yartys VA, Lototsky MV. An overview of hydrogen storage methods. In: *NATO Science Series II: Mathematics, Physics and Chemistry*, 172; 2005. p. 75–104.
- [30] Ioroi T, Kitzawa N, Yasuda K, Yamamoto Y, Takenaka H. IrO₂-deposited Pt electrocatalysts for unitized regenerative polymer electrolyte fuel cells. *J Appl Electrochem* 2001;31:1179–83.
- [31] Nakano A, Ito H, Bhogilla SS, Motyka T, Corgnale C, Greenway S, et al. Research and development for a metal hydride tank with double coil type heat exchanger below 1.0 MPa (G) operation. *Int J Hydrogen Energy* 2015;40:2663–72.